

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 2: Communication

Subtopic 2.1: Networks including the Internet

Concept 2.1.1: LAN vs WAN Characteristics

Total Questions: 7

Mark Schemes begin on Page 5

Question 1 | Source: 9618_w25_qp_12 (Q5.a)

5 A local area network (LAN) has five computers, one switch and one server.

(a) Describe the characteristics of a LAN.

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..... [2]

Question 2 | Source: 9618_s25_qp_12 (Q6.a)

6 A company has multiple sites in different cities.
The company has drivers who deliver products to customers. Each driver can connect to the company's WAN (Wide Area Network) whilst out of the office using a smartphone.

(a) State **two** ways that a WAN is different to a LAN (Local Area Network).

1

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2

..... [2]

Question 3 | Source: 9618_w24_qp_13 (Q9.a)

9 A computer is connected to a Local Area Network (LAN) that connects to a Wide Area Network (WAN).

(a) Describe the characteristics of a WAN.

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..... [2]

Question 4 | Source: 9618_s23_qp_12 (Q1.a)

1 A company has a LAN (local area network).

(a) Give **two** benefits of connecting computers to a LAN.

1

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2

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[2]

Question 5 | Source: 9618_s23_qp_12 (Q1.b)

1 A company has a LAN (local area network).

(b) Give **two** characteristics of a LAN.

1

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2

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[2]

Question 6 | Source: 9618_s23_qp_13 (Q2.a)

2 A university has two sites. Each site has several computer rooms. The computers are all connected as a WAN (wide area network).

(a) Identify **two** differences between a WAN and a LAN (local area network).

1

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2

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[2]

Question 7 | Source: 9618_w21_qp_11 (Q8.a)

8 A school is setting up a network within one of its buildings.

- (a) State whether the network will be a LAN (local area network) or a WAN (wide area network). Justify your choice.

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..... [3]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_12 (Q5.a)

5(a)	1 mark per bullet point, max 2 marks - Covers a small geographical area - Normally privately owned / dedicated infrastructure - Can use wired or wireless communication	2
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Mark Scheme for Question 2 | Source: 9618_s25_ms_12 (Q6.a)

6(a)	1 mark each to max 2 - In a WAN devices can connect over a larger geographical area in a LAN devices connect over a smaller geographical area - A WAN may use non-dedicated hardware the hardware in a LAN is usually dedicated - A WAN uses external / not company owned transmission media // the transmission media for a LAN is usually company owned	2
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Mark Scheme for Question 3 | Source: 9618_w24_ms_13 (Q9.a)

9(a)	1 mark for each bullet point (max 2) - A WAN covers a large geographical area - External/public infrastructure is used - Non-dedicated hardware	2
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Mark Scheme for Question 4 | Source: 9618_s23_ms_12 (Q1.a)

1(a)	1 mark each to max 2 - Allows the sharing of files/data // Allows communication between the devices - Allows the sharing of resources e.g. hardware / software (applications) - Allows central management // by example, backup, security, etc.	2
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Mark Scheme for Question 5 | Source: 9618_s23_ms_12 (Q1.b)

1(b)	1 mark each to max 2 - Covers a small geographical area - The infrastructure is privately owned // not controlled by external organisations	2
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Mark Scheme for Question 6 | Source: 9618_s23_ms_13 (Q2.a)

2(a)	1 mark each to max 2 • WAN covers a large geographical area and LAN covers a small geographical area • LAN connections between devices are usually physical, whereas the WAN connections are often virtual • A LAN has a high data transfer rate, whereas a WAN has a low data transfer rate • The ownership of a LAN is private; the ownership of a WAN can be private or public • LAN is usually more secure than a WAN because protection is easier to implement	2
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Mark Scheme for Question 7 | Source: 9618_w21_ms_11 (Q8.a)

8(a)	1 mark per bullet point • LAN • Small geographical area • No leasing external infrastructure / transmission media // does not use internet to transmit within the building	3
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